

LUCKY LAKE, SK, Canada

WMO: 714550

Lat: 50.950N

Lon: 107.150W

Elev: 2181

StdP: 13.57

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.9	-18.9	-30.8	1.1	-23.8	-25.3	1.5	-18.8	34.5	28.2	30.3	24.7	9.4	220	0.755

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	87.7	64.0	83.9	63.1	80.3	61.7	68.7	79.3	66.3	77.5	64.3	75.4	11.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	101.4	73.8	62.6	92.1	70.6	60.3	84.7	68.5	34.3	78.7	32.2	77.6	30.6	75.3	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.0	25.6	22.9	DB	-29.7	94.7	4.8	3.3	-33.1	97.0	-35.9	99.0	-38.6	100.8	-42.1	103.2	(4)
(5)			WB	-29.9	73.5	4.7	3.5	-33.3	76.1	-36.1	78.2	-38.7	80.1	-42.2	82.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.5	9.7	11.8	23.9	39.7	51.6	59.9	65.2	63.8	55.2	41.1	25.3	13.1		
	DBStd	23.00	15.73	14.49	14.78	10.32	8.41	6.35	5.52	6.34	8.67	9.39	12.82	14.96		
	HDD50	5778	1251	1068	810	330	82	5	0	1	45	300	744	1142		
	HDD65	9850	1716	1488	1275	759	420	176	66	100	311	740	1192	1607		
	CDD50	1581	0	0	1	20	133	302	472	428	199	24	2	0		
	CDD65	179	0	0	0	0	6	23	73	61	16	0	0	0		
	CDH74	2665	0	0	0	12	147	334	893	938	333	8	0	0		
	CDH80	830	0	0	0	2	28	82	276	335	107	0	0	0		
	WSAvg	11.4	12.1	11.4	11.9	12.2	12.5	11.6	9.6	9.5	10.7	11.8	11.8	11.6		
	PrecAvg	15.3	0.7	0.4	0.6	0.9	1.9	3.3	2.4	1.9	1.3	0.7	0.7	0.7		
	PrecMax	21.2	1.6	0.6	1.4	2.4	5.0	8.9	4.9	4.8	3.0	1.9	1.9	1.4		
(15) Precipitation	PrecMin	8.6	0.1	0.1	0.0	0.2	0.2	1.0	0.6	0.2	0.2	0.2	0.1	0.1		
	PrecStd	3.2	0.4	0.2	0.4	0.6	1.1	1.8	1.3	1.2	0.8	0.4	0.5	0.4		
	DB	42.6	42.1	59.0	73.9	83.7	88.2	91.6	93.1	88.7	74.3	59.8	46.4			
	MCWB	37.8	36.9	47.1	51.9	57.7	63.3	66.3	63.4	61.0	55.2	47.1	39.0			
	2% DB	38.3	38.0	52.2	67.4	78.1	81.7	86.7	88.1	82.8	66.3	50.9	40.9			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	34.7	34.3	42.6	49.4	56.0	61.9	66.5	62.8	58.3	50.8	41.8	35.7			
	DB	35.6	35.1	46.3	62.3	73.5	77.5	82.8	83.7	77.2	60.3	45.3	37.4			
	MCWB	32.7	32.4	39.5	46.6	53.6	59.6	66.1	62.2	56.8	48.1	38.6	33.5			
	10% DB	31.7	31.9	40.9	56.9	68.8	73.8	78.9	79.3	71.7	55.1	40.8	33.3			
	MCWB	29.6	30.1	36.6	44.5	51.6	58.5	64.1	61.2	54.7	45.6	36.4	30.4			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	37.7	37.6	47.7	54.0	60.7	69.1	73.9	68.4	63.4	56.6	47.6	39.4			
	MCDB	41.8	41.6	58.3	69.8	74.3	78.8	83.7	81.3	82.1	69.8	59.0	45.0			
	2% WB	35.0	34.6	43.2	50.5	57.6	65.0	70.7	66.5	60.3	52.3	42.6	36.0			
	MCDB	37.9	37.5	50.7	64.1	73.0	75.6	80.7	80.0	77.4	63.3	49.9	40.5			
	5% WB	32.8	32.6	39.5	48.0	55.4	62.5	67.9	64.8	58.1	49.0	39.4	33.6			
(31) Mean Daily Temperature Range	MCDB	35.4	35.0	45.3	60.5	69.7	72.9	78.1	77.9	73.3	58.3	44.6	37.3			
	10% WB	29.5	30.0	36.7	45.3	53.2	60.3	65.7	62.9	55.9	46.5	36.4	30.4			
	MCDB	31.7	32.0	40.9	55.5	66.0	70.2	75.4	75.1	70.1	54.4	40.4	33.3			
	5% DB	16.7	17.2	17.4	21.3	23.5	20.5	23.1	25.1	23.9	19.9	16.2	15.7			
	MCDBR	17.3	18.1	23.1	30.6	30.7	27.9	29.1	33.4	33.4	28.2	22.0	18.0			
(36) Clear-Sky Solar Irradiance	MCWBR	14.7	15.3	14.9	15.8	13.9	12.5	13.8	13.2	14.2	15.1	14.3	14.0			
	5% WB	17.9	17.7	21.5	28.2	26.9	22.7	25.2	27.0	30.2	25.8	20.7	18.1			
	MCWBR	15.7	15.4	14.6	15.4	13.1	12.0	14.0	12.5	13.7	14.7	14.0	14.4			
	taub	0.253	0.250	0.271	0.337	0.355	0.356	0.364	0.373	0.328	0.283	0.253	0.224			
	taud	2.336	2.400	2.500	2.353	2.352	2.404	2.395	2.338	2.445	2.554	2.494	2.423			
(42) All-Sky Solar Radiation	Ebn at Noon	251	288	301	287	285	285	281	272	274	272	254	252			
	Edh at Noon	20	24	27	35	37	36	35	36	29	21	17	15			
	RadAvg	369	685	1119	1523	1855	1919	2074	1667	1223	737	387	274			
(45)	RadStd	39	52	88	125	137	138	122	124	112	93	46	28			

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page